

SATForge Practice Test 7

Full review copy — 147 questions, answer key and reasoning included

Module	Difficulty	Questions	Free response
Section 1, Module 1: Reading and Writing	Standard	27	—
Section 1, Module 2: Reading and Writing	Easy	27	—
Section 1, Module 2: Reading and Writing	Hard	27	—
Section 2, Module 1: Math	Standard	22	3
Section 2, Module 2: Math	Easy	22	3
Section 2, Module 2: Math	Hard	22	3

Nothing in this test is reused. The Math is written from scratch and every answer re-derived with sympy from the question itself. Each stem was checked against all 396 Math questions already live in the bank using a template-similarity measure that ignores the numbers, so a question that reuses an old template with new values is rejected, not just an exact duplicate. The highest remaining similarity is 0.73, against a 0.75 limit. Reading & Writing is drawn from material never used in any published test, and the one remaining gap was authored here.

Difficulty is stamped per question, matching its module — the Test 1/2 convention. Tests 3–6 left every question *MEDIUM* regardless of which module it sat in, which is why their Question Bank difficulty badges and filters are wrong. This test does not repeat that.

Status: not published. This is for review. Nothing has been written to the database.

Section 1, Module 1: Reading and Writing

Standard · 27 questions · 32 minutes

1

CAS / CAS-WV

MEDIUM

Mauricio Drelichman and Hans-Joachim Voth’s analysis of the overall debt and revenue of the government of Philip II (who ruled an empire including Spain and Sicily from 1556 to 1598) found an intriguing ____: although the government regularly defaulted on debt, it ran an even larger overall surplus than did the government of eighteenth-century Britain, which historians consider a model of fiscal virtue.

Which choice completes the text with the most logical and precise word or phrase?

A harbinger

B exemplar

C sanction

D incongruity ✓ correct answer

Why this answer. The colon spells out a mismatch — habitual defaults alongside a bigger surplus than the model of fiscal virtue — which is what 'incongruity' names. 'Exemplar' would make Philip II's government the model rather than the anomaly.

2

CAS / CAS-WV

MEDIUM

Critics initially dismissed the technique as a novelty, but its steady adoption by studios over the following decade ____ that assessment.

Which choice completes the text with the most logical and precise word or phrase?

A confirmed

B undermined ✓ correct answer

C anticipated

D restated

Why this answer. 'But' signals that what followed worked against the dismissal, and steady adoption is evidence the technique was more than a novelty. 'Confirmed' and 'restated' would support the critics.

3

CAS / CAS-WV

MEDIUM

Often, the Nobel Prize in Chemistry is given to a single person, such as Theodore William Richards in 1914. But sometimes the Nobel Committee wants to reward work attributed to two or three individuals, in which case, the award is given _____. For instance, in 2020, Jennifer Doudna was among those awarded for “the development of a method for genome editing.”

Which choice completes the text with the most logical and precise word or phrase?

A retroactively

B jointly ✓ correct answer

C ceremoniously

D reluctantly

Why this answer. The contrast is between one recipient and 'two or three individuals', so the blank must mean shared. 'Jointly' is the only choice about sharing an award.

4

CAS / CAS-WV

MEDIUM

Political blogs with conspicuous ideological alignments became an integral component of US media in the early 2000s. While some commentators lauded this development, asserting that such blogs had a welcome transparency missing from traditional news, less _____ observers countered that such blogs tended to ideological extremes that exacerbated political polarization to problematic levels.

Which choice completes the text with the most logical and precise word or phrase?

A misanthropic

B earnest

C sanguine ✓ correct answer

D recalcitrant

Why this answer. 'Less _____ observers countered' sets these observers against the ones who lauded the development, so the blank must name optimism. 'Sanguine' is the only choice meaning hopeful.

5

CAS / CAS-WV

MEDIUM

Possessing an outstanding collection of public art, Chicago has everything from monumental sculptures like Joan Miró's *Miró's Chicago* at sites like Brunswick Plaza to innovative street art like Justus Roe's mural *South Shore* located on South Exchange Avenue. The _____ public art on display in the city can thus satisfy any art lover.

Which choice completes the text with the most logical and precise word or phrase?

A breadth of ✓ correct answer

B controversy over

C confusion about

D apathy toward

Why this answer. 'Everything from monumental sculptures... to innovative street art' describes range, and 'can thus satisfy any art lover' depends on that range. The other three are negative and unsupported.

6

CAS / CAS-TS

MEDIUM

Researchers César A. Hidalgo, Elisa Castañer, and Andres Sevtsuk created a computer model to predict the mix of businesses and places of interest found in a given neighborhood. The team used data from the Google Places API service to help identify florists, beauty salons, and other businesses and map their locations. This approach has some limits—data from Places API tend to be restricted to places that are customer facing—but the data set nonetheless provides an extremely reliable source to study colocation patterns of neighborhood amenities.

Which choice best describes the function of the underlined portion in the text as a whole?

A It describes the imprecise and subjective nature of neighborhood boundaries.

B It introduces a model that a research team built to evaluate the mix of amenities in urban neighborhoods.

C It emphasizes the potential utility of the team's model.

D It identifies a specific flaw in using a data set about amenities in cities. ✓ correct answer

Why this answer. The underlined clause is the content of 'this approach has some limits' — it names one specific shortcoming of the Places API data. The model itself is introduced in the first sentence, not here, and the utility claim is the clause after the dash.

The following text is from George Marion McClellan’s 1895 poem “Eternity.”

My spirit swoons, and all my senses cry
For Ocean’s breast and covering of the sky.
Rock me to sleep, ye waves, and outward bound,
Just let me drift far out from toil and care,
Where lapping of the waves shall be the sound,
Which mingled with the winds that gently bear
Me on between a peaceful sea and sky,
To make my soothing slumberous lullaby.

Which choice best states the main purpose of the text?

- A** To convey the speaker’s longing for the ocean to impart a sense of inner tranquility ✓ **correct answer**
- B** To contrast the demands of the speaker’s everyday life with the serenity of being rocked to sleep by the ocean
- C** To illustrate the increasing intensity of the speaker’s desire to escape ongoing hardship by gliding on the ocean
- D** To justify the speaker’s qualms about being transported by the ocean to a quiet destination

Why this answer. Every image — ‘rock me to sleep’, ‘peaceful sea and sky’, ‘soothing slumberous lullaby’ — is a wish for calm. ‘Toil and care’ is named once in passing and never developed into the contrast B requires, and the poem expresses no qualms (D).

8

INI / INI-CI

MEDIUM

Eighteenth-century economist Adam Smith is famed for his metaphor of the invisible hand, which he putatively used to illustrate a robust model of how individuals produce aggregate benefits by pursuing their own economic interests. Note “putatively”: as Gavin Kennedy has shown, Smith deploys this metaphor only once in his economic writings—to make a narrow point about the then-dominant economic theory of mercantilism—and it was largely ignored until some twentieth-century economists eager to secure an intellectual pedigree for their views elevated it to a fully-fledged paradigm.

Which choice best states the main idea of the text?

A Some twentieth-century economists gave Smith’s metaphor of the invisible hand a significance it does not have in Smith’s work, but it is nevertheless a useful model of how individuals produce aggregate benefits by pursuing their own economic interests.

B Smith’s metaphor of the invisible hand has been interpreted as a model of how individuals acting in their own interest produce aggregate benefits, but it was intended as a subtle critique of the economic theory of mercantilism.

C The reputation of Smith’s metaphor of the invisible hand is not due to the importance of the metaphor in Smith’s work but rather to the promotion of the metaphor by some later economists for their own ends. ✓ correct answer

D Although Smith is famed for his metaphor of the invisible hand, the metaphor was largely ignored until economists in the twentieth century came to realize that the metaphor was a robust model that anticipated their own views.

Why this answer. ‘Putatively’, ‘only once’, ‘a narrow point’ and ‘largely ignored’ all deflate the metaphor’s role in Smith’s own work, while ‘economists eager to secure an intellectual pedigree for their views’ supplies the motive. A endorses the model the text calls putative, and D says the later economists discovered its merit rather than promoted it for their own ends.

9

INI / INI-CI

MEDIUM

The food industry has long used thermal technologies to preserve food in large batches. Recent advancements in radiofrequency heating were made through research in Mexico on the preservation of wheat flour. Radiofrequency heating is generally considered to be an improvement over more conventional thermal preservation methods: whereas conventional methods transfer energy from the surface of a food to its interior, radiofrequency heating uses precisely controlled radio waves to generate heat within the food itself, thus reducing industrial cooking times.

Based on the text, what is one disadvantage of some conventional thermal preservation methods?

A They require more energy than other methods do.

B Their cooking times are longer than those of other methods. ✓ correct answer

C They are only effective in heating foods that are lightweight and small in size.

D They are especially vulnerable to technological breakdowns.

Why this answer. The only comparison the text draws is that radiofrequency heating reduces cooking times relative to conventional surface-to-interior heating. Energy use, food size and breakdowns are never mentioned.

Some birds that store food recover it months later, and researchers disagree about what guides the search. One proposal is that adult birds rely on landmarks near the cache while juveniles have not yet learned to use them. To test this, an experiment timed how long adults and juveniles took to locate a buried cache, first with nearby landmarks in place and then with those landmarks removed.

Average Time to Locate a Buried Food Cache

Group	Landmarks present (seconds)	Landmarks removed (seconds)
Adults	31	94
Juveniles	38	45

Which choice best describes data from the table that support the proposal?

- A** Juveniles took longer than adults did when landmarks were present.
- B** Both groups took longer to locate the cache once the landmarks had been removed.
- C** Removing the landmarks slowed adults from 31 to 94 seconds but slowed juveniles only from 38 to 45. ✓ correct answer
- D** Adults located the cache faster than juveniles did when landmarks were present.

Why this answer. If adults rely on landmarks and juveniles do not, removing the landmarks should cost adults far more — which is exactly the 63-second penalty against the juveniles' 7. A, B and D are all true of the table but say nothing about who depends on the landmarks.

Scholars cite *One Hundred Years of Solitude*, the 1967 novel by Colombian author Gabriel García Márquez, as a foundational text of magical realism, the Latin American style of fiction in which antirealistic plot devices—often borrowed from the folkloric traditions of Indigenous and colonial societies in the Americas—are deployed in an otherwise realistic mode of representation typical of the modern novel. This style has exerted a decisive influence on authors around the world, including Olga Tokarczuk, whose 1996 novel *Primeval and Other Times* resembles classic magical realist novels in its juxtaposition of literary realism with folklore—namely, that of Poland.

Which quotation from a literary scholar would most directly support the claim in the underlined portion of the text?

A “Like many works in the Latin American magical realist tradition, *Primeval and Other Times* is indebted to antirealistic elements in the folkloric tradition of Poland.”

B “The logic of the realistic plot of *Primeval and Other Times* is repeatedly and productively disrupted by the presence of imagery and situations drawn from Polish folklore.” ✓ **correct answer**

C “While *Primeval and Other Times* alternates between realistic and antirealistic modes of representation, details suggesting the influence of Polish folklore nevertheless occur throughout the novel.”

D “Although Polish folklore clearly informs the style and occasionally antirealistic plot of *Primeval and Other Times*, the novel also shows the inarguable influence of the magical realist tradition of Latin America.”

Why this answer. The claim is a *juxtaposition* — folkloric material set inside an otherwise realistic mode. B has both elements interacting: a realistic plot disrupted by Polish folklore. A names only the folkloric debt, C has the two modes alternating rather than juxtaposed, and D shifts to Latin American influence.

Many films in a language other than English grow to be financially successful over the course of their time in movie theaters in the United States, but some become immediate successes in their opening weekends. A journalist claims that a film's reception can greatly influence its long-term success. Indications that the film is of high quality, such as an Oscar nomination, can dramatically boost public interest in the film and thus its overall earnings, even with a relatively modest opening weekend performance: for example, _____

Highest-Grossing Films in a Language Other than English at US Box Office

Title	Lifetime gross earnings	Opening weekend box office gross earnings	US release date	Director	Oscar nominated?
<i>Instructions Not Included</i>	\$44,467,206	\$7,846,426	August 30, 2013	Eugenio Derbez	No
<i>The Girl Who Played with Fire</i>	\$7,638,241	\$904,998	July 9, 2010	Daniel Alfredson	No
<i>Amélie</i>	\$33,225,499	\$136,470	November 2, 2001	Jean-Pierre Jeunet	Yes
<i>All About My Mother</i>	\$8,272,296	\$50,362	November 5, 1999	Pedro Almodóvar	Yes

Which choice most effectively uses data from the table to complete the example?

A the opening weekend earnings for *Instructions Not Included* were \$7,846,426 and \$904,998 for *The Girl Who Played with Fire*.

B *The Girl Who Played with Fire* earned more money in its opening weekend than *Amélie* earned, but *Amélie* had higher lifetime earnings. ✓ correct answer

C both *Amélie* and *All About My Mother* were recognized by the Oscars, but *Amélie* had higher lifetime earnings.

D *All About My Mother* was recognized by the Oscars, but *Instructions Not Included*, which was not recognized, had higher lifetime earnings.

Why this answer. The example needs a modest opening weekend paired with high lifetime earnings for an Oscar-nominated film. *Amélie* opened at \$136,470 — below *The Girl Who Played with Fire*'s \$904,998 — yet finished at \$33,225,499 against \$7,638,241, and it was the nominated one. A gives opening figures with no lifetime comparison, C compares two nominated films so the nomination explains nothing, and D runs against the claim.

Evan MacLean and colleagues evaluated behavioral and genetic data from over 14,000 dogs, representing more than 100 breeds, and found that variations in behavior between breeds can be attributed to genetic variations between those breeds, suggesting a genetic basis for breed differences in behavior. This was the case for both separation problems and energy but was especially pronounced for trainability, which can be seen when a dog demonstrates willingness to fetch objects. In a different study, researchers found that, with regard to trainability, the English mastiff behaves in notably different ways than the Yorkshire terrier. Together, these findings imply that _____

Which choice most logically completes the text?

- A the English mastiff and the Yorkshire terrier will likely become more genetically similar over time.
- B the English mastiff and the Yorkshire terrier differ with respect to the genetic underpinnings for trainability. ✓**
correct answer
- C individual English mastiffs may display higher levels of trainability than individual Yorkshire terriers.
- D English mastiffs and Yorkshire terriers show a greater tendency toward trainability than most other dog breeds do.

Why this answer. Premise one: between-breed behavioural differences are genetic, especially for trainability. Premise two: these two breeds differ in trainability. The conclusion that follows is a genetic difference in trainability. C picks a direction the text never gives, and A and D go well beyond both premises.

14

INI / INI-IE

MEDIUM

For its 1974 work *Instant Mural*, the Chicano art collective Asco taped members Patssi Valdez and Humberto Sandoval to an outdoor wall in East Los Angeles. The work is manifestly a commentary on constraint, but many critics focus on Valdez and the social constraints women faced at the time, which is understandable but leaves the presence of Valdez's male collaborator Sandoval unexplained. We should instead consider that in 1974, the art establishment's recognition of Chicano artists was (and had long been) restricted to sociohistorical muralists, leaving nonmuralist Chicano artists—like Asco's members—struggling to even exhibit their work; attending to this context opens an interpretation that accounts for all the evidence, allowing us to conclude that _____

Which choice most logically completes the text?

- A** while Valdez's presence in *Instant Mural* represents the social constraints placed on women at the time, Sandoval's presence represents Chicano muralists' frustration at their lack of recognition by the art establishment.
- B** *Instant Mural* is best understood not as a critique of the social constraints placed on women but rather as a critique of sociohistorical muralists' depictions of Chicano culture.
- C** the main subject of *Instant Mural* is female Chicano artists' experience of being doubly constrained by gender-role expectations and the marginalization of certain types of art.
- D** *Instant Mural* is a reflection on the constraining aesthetic expectations placed on Chicano artists in general rather than on the social constraints placed on women specifically. ✓ correct answer

Why this answer. The interpretation must 'account for all the evidence', i.e. explain Sandoval as well as Valdez. Constraint on Chicano artists generally covers both taped figures. A leaves them with unrelated meanings and gets the muralists backwards (it was the muralists who *were* recognised), and C keeps the female-only focus the passage rejects.

15

SEC / SEC-BS

MEDIUM

Icelandic turf houses were built with thick walls of stacked sod to hold heat through the winter, _____ the same walls kept interiors cool during the long days of summer.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A** and ✓ correct answer

B and,

C however

D therefore

Why this answer. Both halves are independent clauses, so the comma needs a coordinating conjunction after it. 'However' and 'therefore' are conjunctive adverbs, which cannot join clauses with only a comma, and B adds a comma with nothing to set off.

16

SEC / SEC-BS

MEDIUM

A traditional Japanese toolbox holds only a handful of implements—a pull saw, a marking gauge, a set of chisels, and a _____ each of which is sharpened rather than replaced.

Which choice completes the text so that it conforms to the conventions of Standard English?

A plane— ✓ **correct answer**

B plane,

C plane;

D plane

Why this answer. A dash opened the list, so a dash closes it before the relative clause 'each of which...'. A comma or semicolon would leave the opening dash unmatched.

17

SEC / SEC-BS

MEDIUM

The axolotl—a salamander that keeps its larval features into _____ can regrow not only its limbs but also parts of its heart and brain.

Which choice completes the text so that it conforms to the conventions of Standard English?

A adulthood,

B adulthood

C adulthood— ✓ **correct answer**

D adulthood:

Why this answer. A dash opens the supplementary description of the axolotl, so a dash must close it. Swapping in a comma or a colon breaks the pair, and omitting the mark leaves the aside running into the verb.

Despite the fact that what is now known as the Cassegrain reflector was first constructed by Bonaventura Cavalieri, the telescope is named after the seventeenth-century French astronomer Laurent Cassegrain. One might assume cases like this, where a discovery or concept is not named after the first person who discovered it, to be _____ they are counted among a litany of examples of Stigler's law.

Which choice completes the text so that it conforms to the conventions of Standard English?

A rare, on the contrary,

B rare, on the contrary;

C rare on the contrary,

D rare; on the contrary, ✓ correct answer

Why this answer. 'One might assume... to be rare' and 'they are counted among a litany of examples' are both independent clauses, and 'on the contrary' is a conjunctive adverb opening the second. That needs a semicolon before it and a comma after — A and C are comma splices, and B strands 'on the contrary' at the end of the first clause.

Most US states and territories have had at least one woman serve as governor. In Nebraska, for example, Governor Kay A. Orr took office on January 9, 1987. In fact, the number of states and territories that have had only male governors _____ almost every year.

Which choice completes the text so that it conforms to the conventions of Standard English?

A shrinks ✓ correct answer

B shrink

C are shrinking

D have been shrinking

Why this answer. The subject is the singular 'the number' (not the plural 'states and territories' inside the modifier), and 'almost every year' calls for the simple present. Only 'shrinks' is both singular and simple present.

The Suzuki method is an approach to music instruction _____ to teach music in the same natural way children learn language, emphasizes retaining and reviewing previously learned pieces of music. In Suzuki piano instruction, students begin their musical education with simple folk songs like “The Honeybee.”

Which choice completes the text so that it conforms to the conventions of Standard English?

A that, is aiming

B that, aiming ✓ correct answer

C that aims

D that has aimed

Why this answer. The relative clause's verb is 'emphasizes', which follows the comma after 'language', so the blank must open a parenthetical participial phrase: 'that, aiming to teach music..., emphasizes...'. Any finite verb in the blank leaves 'emphasizes' stranded without a subject.

The weavers guild kept its patterns secret for generations, and even today several of the _____ techniques are known only to families who have practiced them since the 1700s.

Which choice completes the text so that it conforms to the conventions of Standard English?

A guilds

B guild's ✓ correct answer

C guilds'

D guilds's

Why this answer. The techniques belong to the one guild named in the first clause, so the singular possessive is correct. A is a bare plural, C is a plural possessive, and D is not a form English uses.

22

EOI / EOI-TR

MEDIUM

The work of contemporary Asian American poet John Yau often incorporates references to other poems and works of visual art. Typically, these allusive gestures are subtle, recognized by only the most attentive of Yau’s readers. In his 2013 poem “Further Adventures in Monochrome,” _____ Yau explicitly identifies a particular artwork—Yves Klein’s 1960 painting *Silence is Golden*—as the poem’s inspiration.

Which choice completes the text with the most logical transition?

A though, ✓ correct answer

B fittingly,

C similarly,

D for example,

Why this answer. ‘Explicitly identifies’ is the opposite of the ‘typically... subtle’ allusions just described, so the 2013 poem is an exception, not an illustration (D) or a parallel case (C).

23

EOI / EOI-TR

MEDIUM

Many English adjectives share a linguistic origin with their associated nouns, like the adjective “monetary” and the noun “money,” both of which come from the Latin “monetarius.” _____ some adjectives do not share an origin with their associated nouns. For example, the adjective “bovine” ultimately comes from the Latin “bos,” while its associated noun, “cow,” comes from the Old English “cū.”

Which choice completes the text with the most logical transition?

A Subsequently,

B For this reason,

C Nevertheless, ✓ correct answer

D Specifically,

Why this answer. ‘Some adjectives do not share an origin’ contradicts the opening generalisation, so a contrast is needed. The other three signal sequence, cause and elaboration.

In Argentina, the Chamber of Deputies is elected via a proportional representation (PR) system. In PR elections, votes are cast (not for specific candidates, as they are in single-member plurality systems, but for political parties) and then tabulated; _____ each qualifying party is awarded a number of seats proportional to the number of votes it received.

Which choice completes the text with the most logical transition?

A second of all,

B in fact,

C by contrast,

D accordingly, ✓ correct answer

Why this answer. Awarding seats in proportion to the votes counted is the consequence of the tabulation just described. 'Second of all' needs a first, there is no contrast with the preceding clause (C), and nothing is being intensified (B).

While researching a topic, a student has taken the following notes:

- Dinosaur fossil specimens can be found at science museums all over the world.
- Many dinosaur fossil specimens are given nicknames.
- The Great Plains Dinosaur Museum and Field Station in Malta, Montana, houses a dinosaur fossil specimen nicknamed Roberta.
- Roberta lived in the Late Cretaceous period, which ended more than 65 million years ago.
- It is a member of the genus *Brachylophosaurus*.

The student wants to provide an example of a dinosaur fossil specimen's nickname. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A Nicknames are given to many dinosaur fossil specimens, including one housed at a museum in Malta, Montana.

B Dinosaur fossil specimens can be found at museums all over the world, and many of these specimens are given nicknames.

C Roberta is the nickname of a *Brachylophosaurus* fossil specimen housed at the Great Plains Dinosaur Museum and Field Station in Malta, Montana. ✓ correct answer

D A *Brachylophosaurus* fossil specimen from the Late Cretaceous period, which ended more than 65 million years ago, is housed at the Great Plains Dinosaur Museum and Field Station.

Why this answer. Only C actually states a nickname ('Roberta'). A and B say nicknames exist without giving one, and D omits the nickname entirely.

While researching a topic, a student has taken the following notes:

- Louis Ballard was a classical composer and citizen of the Quapaw Tribe.
- His compositions synthesize Western classical music with elements of various Native musical traditions.
- Ballard's classical piece *Katcina Dances* features traditional Hopi songs.
- Ballard's classical piece *Fantasy Aborigine No. 3* incorporates a Tewa seashell rattle.

The student wants to emphasize a difference between the two compositions. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A *Katcina Dances* and *Fantasy Aborigine No. 3* are two compositions written by Ballard, a classical music composer.

B In his two compositions *Katcina Dances* and *Fantasy Aborigine No. 3*, Ballard blends elements of various Native musical traditions, such as Indigenous songs or instruments, with Western classical music.

C Ballard has different approaches to blending Western classical music with elements of various Native musical traditions, such as using Indigenous songs and instruments in his compositions.

D While both compositions integrate various Native musical traditions, *Katcina Dances* does so by featuring traditional Hopi songs and *Fantasy Aborigine No. 3* does so by incorporating a Tewa seashell rattle. ✓ correct answer

Why this answer. D names what each piece does differently — Hopi songs versus a Tewa seashell rattle — against a shared background. A and B stress what the two have in common, and C asserts a difference without saying what it is.

While researching a topic, a student has taken the following notes:

- Ixmiquilpan is a municipality in the state of Hidalgo, Mexico.
- Municipalities are governmental regions responsible for providing many public services to their residents.
- One service they provide is water treatment.
- Ixmiquilpan's population was 98,654 in 2020.
- Hidalgo is divided into 84 municipalities.

The student wants to provide an example of a public service that Ixmiquilpan is responsible for. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A As a municipality, Ixmiquilpan is responsible for providing water treatment to its residents. ✓ correct answer

B Ixmiquilpan is one of 84 municipalities in Hidalgo providing public services to their communities.

C Ixmiquilpan—a governmental region in the state of Hidalgo, Mexico—provides public services to its residents.

D In 2020, the municipality of Ixmiquilpan had a population of 98,654.

Why this answer. Only A names a specific service (water treatment). B and C stay at the general 'public services' level and D is about population.

Section 1, Module 2: Reading and Writing

Easy · 27 questions · 32 minutes

1

CAS / CAS-WV

EASY

The composer's late works are unusually ____: a single movement may quote a folk tune, a hymn, and a march within the space of two minutes.

Which choice completes the text with the most logical and precise word or phrase?

A restrained

B eclectic ✓ correct answer

C repetitive

D somber

Why this answer. Drawing on a folk tune, a hymn and a march at once is drawing from varied sources. 'Repetitive' would require the same material to return, which is the opposite of what the example shows.

2

CAS / CAS-WV

EASY

Some social scientists say that while an emphasis on preserving civil liberties is key to democracy, public understanding of history is also central to public comprehension of state politics, and if a citizenry is to function, historical issues cannot remain the domain only of experts. In short, knowledge of history is not ____ and must not be left to historians alone.

Which choice completes the text with the most logical and precise word or phrase?

A unattainable

B superfluous ✓ correct answer

C commonplace

D equitable

Why this answer. The summary restates that historical knowledge is 'central' and belongs to everyone, so 'not ____' must deny that it is unnecessary. 'Not unattainable' and 'not commonplace' would say nothing about its importance.

3

CAS / CAS-WV

EASY

Cybersecurity experts often encourage users to create passwords that are fairly complicated and therefore difficult to guess. Nonetheless, research has shown that the more _____ approach to password selection seems to favor convenience over security: for example, the eighth most commonly used password in 2019 was the easily remembered “iloveyou.”

Which choice completes the text with the most logical and precise word or phrase?

A creative

B popular ✓ correct answer

C complex

D useful

Why this answer. The example is about what people most commonly choose ('the eighth most commonly used password'), so the blank names prevalence. 'Complex' would contradict 'favor convenience over security'.

4

CAS / CAS-WV

EASY

The swordfish can swim very fast—up to 97 kilometers per hour (km/hr)—but it is significantly slower than the golden eagle, which can fly at speeds up to 320 km/hr. The difference between these speeds is largely _____ of the fact that the features that make flight possible do less to limit top speeds than the features suitable for swimming through water.

Which choice completes the text with the most logical and precise word or phrase?

A a consequence ✓ correct answer

B an objective

C a repudiation

D an explanation

Why this answer. The stated fact about flight versus swimming is the cause and the speed gap is the effect, so the gap is 'a consequence of' it. 'An explanation of' reverses which one explains which.

5

CAS / CAS-WV

EASY

In a 2018 article about films depicting the experiences of Black Americans, critics for the *New York Times* praise Madeline Anderson’s 1970 film *I Am Somebody* as “galvanizing” and Reginald Hudlin’s 1990 film *House Party* as “exuberant.” Fans of the two films hope that such _____ will attract new audiences to these works.

Which choice completes the text with the most logical and precise word or phrase?

A impartiality

B ambivalence

C foresight

D acclaim ✓ correct answer

Why this answer. 'Such _____' refers back to the critics' praise. Only 'acclaim' names praise; 'impartiality' and 'ambivalence' contradict it.

6

CAS / CAS-TS

EASY

Scholarly interest in literary juvenilia—writings by children and teenagers—tends to focus on unpublished works by authors who became famous as adults, such as W.H. Auden’s poem “Autumn,” which he wrote around age 15, because they offer insights into their authors’ artistic development. But some scholars also argue that recovering juvenilia by lesser-known writers is essential to understanding literary history: Daisy Ashford’s novels, which she published as a child, were widely read by contemporaries and are therefore deserving of closer attention.

Which choice best states the main purpose of the text?

A To describe the challenges famous writers encountered when seeking to publish works written in their childhood

B To argue that Ashford’s novels have more literary merit than Auden’s juvenilia do

C To compare the accomplishments of young writers with those of their adult contemporaries

D To present reasons why literary scholars consider juvenilia to be valuable resources ✓ correct answer

Why this answer. The text gives two reasons juvenilia is studied — insight into artistic development, and understanding literary history — without ranking the two writers or discussing publication difficulties.

During Rome’s republican period, which ended in the first century BCE, libraries were predominantly owned by wealthy individuals who tightly controlled access to their book collections. The first public library became available in Rome in 28 BCE and was soon followed by one commissioned by Emperor Augustus. As modern scholar Fabio Fernandes notes, however, these two traditions aren’t as distinct as they seem, as both the emperor and the private library owners viewed their libraries as extensions of their personal patronage, just on vastly differing scales.

Which choice best states the main purpose of the text?

- A To give a brief overview of public access to libraries throughout Rome’s republican period
- B To contend that early imperial leaders in Rome wielded too much influence over libraries
- C To assert that private and early public libraries in ancient Rome had an essential similarity ✓ correct answer
- D To call into question the notion that private Roman libraries disappeared during the first century BCE

Why this answer. The text builds to Fernandes's point that the two traditions 'aren't as distinct as they seem' because both served personal patronage. That is a claim of similarity, not a critique of imperial influence (B) or a survey of public access (A).

The following text is from George Eliot’s 1857 short story “The Sad Fortunes of the Rev. Amos Barton.” In the text, the narrator addresses the reader directly and alludes to a discussion among Rev. Amos Barton’s neighbors.

It was happy for the Rev. Amos Barton that he did not, like us, overhear the conversation recorded in the last chapter. Indeed, what mortal is there of us, who would find his satisfaction enhanced by an opportunity of comparing the picture he presents to himself of his own doings, with the picture they make on the mental retina of his neighbours? We are poor plants buoyed up by the air-vessels of our own conceit: alas for us, if we get a few pinches that empty us of that windy self-subsistence! The very capacity for good would go out of us.

Which choice best states the main idea of the text?

- A Although people grasp the importance of honesty, they typically resist confronting others about their flaws.
- B Although people wish to be seen as considerate, the slightest setbacks will often discourage them from being so.
- C People tend to fixate more often than they should on whether their acquaintances think highly of them.
- D People are better off not knowing about the discrepancy between their own self-image and what others think of them. ✓ correct answer

Why this answer. 'It was happy for [Barton] that he did not overhear' plus the rhetorical question about comparing one's self-image with 'the picture they make on the mental retina of his neighbours' — the narrator argues such knowledge would drain the 'capacity for good'.

The following text is from Virginia Woolf's 1919 novel *Night and Day*. The narrator describes a gathering of artists and intellectuals.

One person after another rose, and, as with an ill-balanced axe, attempted to hew out his conception of art a little more clearly, and sat down with the feeling that, for some reason which he could not grasp, his strokes had gone awry. As they sat down they turned almost invariably to the person sitting next them, and rectified and continued what they had just said in public.

The text makes which point about the people at the gathering?

- A Each enjoys speaking publicly about abstract issues but dislikes speaking privately about them.
- B Each fails at presenting a wholly coherent vision of art but does not understand why. ✓ correct answer**
- C Each is contemptuous of the other attendees but strives to impress them.
- D Each becomes absorbed in a question about art, and no one knows how to answer the question.

Why this answer. 'His strokes had gone awry' is the failure and 'for some reason which he could not grasp' is the incomprehension — B's two halves exactly. A reverses the passage, which shows them continuing to talk privately, and nothing suggests contempt (C).

Water flowing around an obstruction creates vortices (patterns of swirls) of varying size; by detecting the vortices, fish can determine the size and position of the obstruction. Testing by Yuzo R. Yanagisuru, Otar Akanyeti, and James C. Liao using models of three head shapes—narrow (low ratio of width to length), intermediate, and wide (high ratio of width to length)—showed that for medium-sized vortices, fish with wide heads would be least able to distinguish between vortices and general turbulence in the water. A second research team has therefore hypothesized that in low-visibility conditions, wider-headed fish will be less likely than narrower-headed fish to detect obstructions.

Which finding, if true, would most directly support the second research team’s hypothesis?

A A study using obstructions that created medium-sized vortices in low-visibility conditions found that some specimens of dusky smooth-hound (*Mustelus canis*), which has a relatively narrow head, bumped into the obstructions more often than other specimens of the same fish did.

B A study using obstructions that created medium-sized vortices in low-visibility conditions found that the wider-headed bristlemouth (*Chaetostoma yurubiense*) bumped into obstructions more often than the narrower-headed dusky smooth-hound (*Mustelus canis*) did. ✓ correct answer

C A study using obstructions that created medium-sized vortices in low-visibility conditions found that the narrower-headed dusky smooth-hound (*Mustelus canis*) bumped into the obstructions just as often as the wider-headed bristlemouth (*Chaetostoma yurubiense*) did.

D A study using obstructions that created medium-sized vortices in low-visibility conditions found that the bristlemouth (*Chaetostoma yurubiense*), which has a relatively wide head, bumped into more than half of the obstructions.

Why this answer. The hypothesis is comparative — wider-headed fish detect obstructions less well than narrower-headed ones — so the supporting finding must compare the two head shapes and favour the narrow one. A compares members of one narrow-headed species, C finds no difference, and D reports a wide-headed rate with nothing to compare it against.

Drosophila (fruit flies) have generation times of 10–12 days, so seasonal changes in humidity and other environmental conditions can drive seasonal fluctuations in chromosome rearrangements in species such as *D. persimilis* and *D. mediopunctata*. *Drosophila* body size (for which wing centroid size serves as a proxy measure) correlates with life span. Banu Şebnem Önder and Cansu Fidan Aksoy measured the wing sizes of members of a *D. melanogaster* population in Yeşilöz, Turkey, that were collected monthly between May and October over three years. Their research suggests that *Drosophila* collected in relatively warmer months should tend to have a longer life span, as is illustrated by the finding that _____

Monthly Temperatures and Wing Centroid Sizes of Fruit Fly Specimens

Month	Average high (°F)	Average low (°F)	Average male wing centroid size (mm)	Average female wing centroid size (mm)
October	67	44	1.98	2.29
July	87	62	2.02	2.31
June	80	56	2.01	2.31
May	73	50	1.98	2.27

Which choice most effectively uses data from the table to complete the assertion?

A the average monthly low temperature was higher in June than in May.

B the average male wing centroid size was larger in July than in October. ✓ correct answer

C the average female wing centroid size was 2.02 mm in July but was 2.31 mm in June.

D the average female wing centroid size was consistently larger than the average male wing centroid size in all four months in the table.

Why this answer. The assertion needs warmer month -> larger body size (wing centroid) -> longer life span. July (87/62) is the warmest month and October (67/44) the coolest; male centroid 2.02 vs 1.98 makes exactly that link. A cites temperature with no size, D cites size with no temperature, and C misreads the table (July female is 2.31, not 2.02).

In the 1920s the archaeologist Dorothy Garrod excavated a series of caves at Mount Carmel, in what is now Israel. Garrod employed an almost entirely female workforce drawn from nearby villages, an unusual arrangement for the period. Some later commentators have characterised this choice as a purely practical response to a shortage of available male labourers. A researcher studying Garrod's field notebooks, however, has argued that Garrod deliberately recruited and trained the women because she judged them better suited to the delicate work of clearing fragile deposits.

Which quotation from Garrod's notebooks, if genuine, would most directly support the researcher's argument?

A "The men of the district being occupied with the harvest, I was obliged to look elsewhere for hands."

B "I have set the women to the sieving, having found their touch far surer than any I could teach a labourer, and mean to keep them at it." ✓ correct answer

C "The season has been fine, and the trenches are dry enough that work has gone forward without interruption."

D "Wages here are low, and I am able to engage a great many more hands than the budget would allow in England."

Why this answer. The researcher's claim has two parts: the recruitment was deliberate, and the reason was the women's suitability for delicate work. B carries both — 'having found their touch far surer' gives the judgement of suitability, and 'mean to keep them at it' makes it a choice rather than a necessity. A supports the opposing 'shortage of men' reading. C is about weather and says nothing about the workforce. D explains the size of the workforce by cost, not the choice of women by skill.

Liverpool Football Club won more than half its home matches between 1947 and 2003, helping to distinguish the team as a top British professional soccer club. Additionally, Liverpool, whose home uniform color is mainly red, was more successful in home matches than was Watford Football Club, whose home uniform color is not red. Research has shown that many people associate the color red with power and dominance. Past experiences with red objects such as stop signs can even cause people to respond to the color red with caution and hesitation. These findings, if applicable in the context of athletic competitions, could suggest that in matches played at Liverpool's home stadium, opponents may have _____

Which choice most logically completes the text?

A played more aggressively than they would have if their own uniforms were red.

B been influenced by the color of Liverpool's uniforms. ✓ correct answer

C been aware that Liverpool's uniform color affected their own performance.

D been unaffected by the red clothing worn by fans in the stadium.

Why this answer. Red signals dominance and prompts caution, and at Liverpool's home stadium the red belongs to Liverpool's uniforms — so the inference is about opponents being affected by them. A reverses the effect, C adds awareness the research never claims (the effect is described as automatic), and D switches to the fans' clothing.

14

INI / INI-IE

EASY

A road was recently built in a Maryland woodland that is home to the house finch. Some finches' nests were situated close to the new road and others were deeper in the woodland. Common ravens, which eat finch eggs, can spot nests near the open spaces of roads more easily than they can spot nests surrounded by woodland. Accordingly, researchers in Maryland trying to predict the impact of the new road on finches have suggested that _____

Which choice most logically completes the text?

- A finch eggs will make up about the same percentage of common ravens' diet as they did before the road was built.
- B the number of finches building nests near the road will gradually increase as the finches adapt to the presence of the common ravens.
- C finches nesting near the road will lose eggs to common ravens at a higher rate than will finches nesting far from the road. ✓ correct answer
- D finches nesting far from the road will lay fewer eggs per nest than will finches nesting near the road.

Why this answer. Ravens spot roadside nests more easily and eat finch eggs, so roadside nests should lose more eggs. Nothing in the text supports claims about diet percentage, adaptation, or clutch size.

15

SEC / SEC-BS

EASY

On February 1, 2018, a Florida-based research team—Martha A. Scholl, Maoya Bassiouni, and Angel J. Torres-Sánchez—compiled climate data from several sites in Puerto Rico's Luquillo Mountains. At 8:30 a.m., the air temperature was 16°C at site CC1, the site with the highest _____ it had shifted to 16.8°C by 11:00 p.m.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A elevation and
- B elevation
- C elevation,
- D elevation, and ✓ correct answer

Why this answer. 'The air temperature was 16°C at site CC1, the site with the highest elevation' and 'it had shifted to 16.8°C by 11:00 p.m.' are both independent clauses, so they need a comma plus a coordinating conjunction. C alone is a comma splice and B is a run-on.

16

SEC / SEC-BS

EASY

Featuring works by the photographers Liselotte Grschebina and Else “Yva” Neuländer-Simon, the 2021 exhibition *The New Woman Behind the Camera* set out to provide a wide-ranging overview of photography by women in the 1920s through the ____ given the collection’s breadth of more than 120 photos, its efforts were largely successful.

Which choice completes the text so that it conforms to the conventions of Standard English?

A 1950s,

B 1950s

C 1950s and

D 1950s, and ✓ correct answer

Why this answer. Two independent clauses ('the exhibition set out to provide...' and 'its efforts were largely successful') need a comma plus a coordinating conjunction. A alone is a comma splice, B is a run-on, and C drops the comma required before 'and' joining independent clauses.

17

SEC / SEC-BS

EASY

Moussa Ould Ebnou is a novelist from Boutilimit, Mauritania, who has earned international acclaim in recent years. Writers of other literary forms, such as poetry and drama, are less likely to be recognized beyond their nations’ borders, but many are still well ____ Koulsy Lamko, for example, is a celebrated playwright from Chad.

Which choice completes the text so that it conforms to the conventions of Standard English?

A known,

B known: ✓ correct answer

C known and

D known

Why this answer. The second clause is a specific illustration of 'many are still well known', which a colon introduces. A is a comma splice, C joins two independent clauses without a comma, and D leaves a run-on.

18

SEC / SEC-BS

EASY

Bioluminescent fungi glow steadily rather than in _____ researchers studying one species recorded emission that varied by less than a tenth over a full night.

Which choice completes the text so that it conforms to the conventions of Standard English?

A pulses,

B pulses. Researchers ✓ correct answer

C pulses, researchers

D pulses researchers

Why this answer. Two complete thoughts require a sentence boundary. A leaves the second clause without a subject position, C is a comma splice, and D is a run-on. Only B closes the first sentence and starts the second.

19

SEC / SEC-FS

EASY

Trade was vital to the Srivijaya Empire, which reigned in Southeast Asia from around 600 CE to 1200 CE. Its people _____ tin, medicines, and wood to sell to neighboring societies. In exchange, they received valuable items, such as spices, silk, and porcelain.

Which choice completes the text so that it conforms to the conventions of Standard English?

A are producing

B produce

C produced ✓ correct answer

D will produce

Why this answer. The surrounding sentences are in the simple past ('was vital', 'they received') about an empire that ended in 1200 CE, so the past tense 'produced' is required.

20

SEC / SEC-FS

EASY

Among the instruments recovered from the shipwreck _____ a brass astrolabe engraved with a date two decades earlier than the voyage itself.

Which choice completes the text so that it conforms to the conventions of Standard English?

A were

B was ✓ correct answer

C have been

D are

Why this answer. The sentence is inverted: the subject is the singular 'a brass astrolabe', which follows the verb. 'Instruments' sits inside the opening prepositional phrase and cannot govern the verb.

21

SEC / SEC-FS

EASY

Cartographers _____ the coastline before satellite imagery existed relied on chains of triangles measured from hilltop to hilltop.

Which choice completes the text so that it conforms to the conventions of Standard English?

A mapping ✓ correct answer

B mapped

C map

D have mapped

Why this answer. 'Relied' is already the sentence's main verb, so the blank must open a participial phrase modifying 'cartographers'. Any finite form gives the subject two verbs with no conjunction.

22

EOI / EOI-TR

EASY

To cast a bronze sculpture by the lost-wax method, an artist first builds a wax model and coats it in a heat-resistant shell. _____ the wax is melted out, leaving a cavity that molten bronze will fill.

Which choice completes the text with the most logical transition?

A In summary,

B Next, ✓ correct answer

C By comparison,

D Regardless,

Why this answer. 'First' in the preceding sentence sets up a sequence of steps, and melting the wax is the step that follows.

23

EOI / EOI-TR

EASY

The decades since the Second World War have seen a range of outcomes for the independence movements of Micronesia, Melanesia, and Polynesia. Many of the regions' islands and groups of islands have become independent nations. _____ the Northern Mariana Islands, a commonwealth in Micronesia consisting of fifteen islands, are part of the US, and the Society Islands, a group of islands in Polynesia including Tahiti and Bora Bora, are part of the territory of French Polynesia.

Which choice completes the text with the most logical transition?

A For example,

B Therefore,

C In fact,

D On the other hand, ✓ correct answer

Why this answer. Islands that remain part of the US and of French Polynesia are the opposite of the preceding sentence's independent nations — the 'range of outcomes' the first sentence promised. They are not examples of independence (A), nor a consequence (B) or intensification (C) of it.

24

EOI / EOI-TR

EASY

Several languages mark evidentiality, requiring a speaker to indicate how a claim was learned. _____ in Tariana a verb ending distinguishes something seen from something merely reported by another person.

Which choice completes the text with the most logical transition?

A For example, ✓ **correct answer**

B However,

C Instead,

D Therefore,

Why this answer. Tariana is one instance of the general claim about evidentiality, not a contrast with it or a consequence of it.

25

EOI / EOI-RS

EASY

While researching a topic, a student has taken the following notes:

- The A.M. Turing Award is a prestigious award given for “major contributions of lasting importance to computing.”
- Manuel Blum won the award in 1995 for contributions to the foundations of computational complexity theory.
- Raj Reddy won the award in 1994 for pioneering the development of large-scale artificial intelligence systems.

The student wants to emphasize the order in which Manuel Blum and Raj Reddy won the A.M. Turing Award. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A Manuel Blum and Raj Reddy both won the A.M. Turing Award, which is given for “major contributions of lasting importance to computing.”

B Raj Reddy won the A.M. Turing Award in 1994; Manuel Blum won it later, in 1995. ✓ **correct answer**

C In 1995, Manuel Blum won the A.M. Turing Award for contributions to the foundations of computational complexity theory.

D It was in 1994 that Raj Reddy won the A.M. Turing Award.

Why this answer. Only B mentions both men and marks the sequence explicitly with 'later'. C and D each cover one winner, and A gives no dates at all.

While researching a topic, a student has taken the following notes:

- A qanat is an underground channel that carries water from an aquifer to the surface by gravity alone.
- Qanats have been built across arid regions of Asia and North Africa for more than 2,500 years.
- The Qanat of Gonabad in Iran is about 33 kilometers long.
- Its main well shaft reaches a depth of roughly 300 meters.
- It still supplies water to farms today.

The student wants to emphasize the depth of the Qanat of Gonabad's main well shaft. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A The Qanat of Gonabad, which is about 33 kilometers long, is an underground channel in Iran.

B Qanats have carried water across arid regions of Asia and North Africa for more than 2,500 years.

C The main well shaft of the Qanat of Gonabad descends roughly 300 meters below the surface. ✓ correct answer

D The Qanat of Gonabad still supplies water to farms today, using gravity alone.

Why this answer. Only the 'descends roughly 300 meters' choice gives the shaft depth. One choice gives the 33-kilometer length instead, and the other two are about qanats generally and about present-day use.

While researching a topic, a student has taken the following notes:

- Square Tower House is a former Ancestral Puebloan settlement in southwestern Colorado that was inhabited from approximately 1200–1300 CE.
- It contained elaborate multistory buildings primarily made of adobe and stone.
- The Ancestral Puebloan civilization (approximately 1200 BCE–1600 CE) was the precursor of modern Pueblo tribal nations.
- These modern nations include Zuni Pueblo and Isleta Pueblo.
- Buildings in Zuni Pueblo and Isleta Pueblo often incorporate traditional elements, such as adobe and stone, alongside more contemporary elements.

The student wants to connect Square Tower House to modern-day Pueblo buildings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A Modern-day Pueblo buildings, such as those in Zuni Pueblo and Isleta Pueblo, incorporate elements that wouldn't have been found in Square Tower House.

B The Ancestral Puebloan settlement of Square Tower House was built with adobe and stone, materials still used in the buildings of Zuni Pueblo and Isleta Pueblo. ✓ **correct answer**

C The elaborate buildings of Square Tower House in southwestern Colorado incorporate both traditional and modern-day elements in their design.

D Square Tower House, located in southwestern Colorado, was inhabited from approximately 1200–1300 CE, but its buildings aren't actively occupied today.

Why this answer. B is the only choice naming something the two share — adobe and stone. A stresses a difference, C misattributes modern elements to Square Tower House, and D mentions only Square Tower House.

Section 1, Module 2: Reading and Writing

Hard · 27 questions · 32 minutes

1

CAS / CAS-WV

HARD

The following text is adapted from the 1895 poem “Ojistoh” by Emily Pauline Johnson, a Kaniienkahagen (Mohawk) writer also known as Tekahionwake.

I am Ojistoh, I am she, the wife
Of him whose name breathes bravery and life
And courage to the tribe who calls him chief.
I am Ojistoh, his white star, and he
Is land, and lake, and sky—and soul to me.

As used in the text, what does the word “breathes” most nearly mean?

A Imparts ✓ correct answer

B Renounces

C Assents

D Absorbs

Why this answer. His name gives bravery, life and courage *to* the tribe, so the verb must mean to give out. 'Absorbs' reverses the direction and the other two do not fit.

2

CAS / CAS-WV

HARD

Because seed banks must remain viable for decades without power, engineers site them in permafrost, where the surrounding ground keeps temperatures low even if the cooling system fails—a design that is deliberately _____.

Which choice completes the text with the most logical and precise word or phrase?

A redundant ✓ correct answer

B temporary

C compact

D ornamental

Why this answer. Ground that holds the temperature when the cooling system fails is a backup for a function already provided, which is what a redundant design means. The others describe duration, size and appearance.

3

CAS / CAS-WV

HARD

Museums once treated conservation as a matter of arresting decay, and any change to an object was considered a loss. Newer approaches temper that view, accepting that some materials were designed to age and that halting the process can misrepresent what the maker intended.

As used in the text, what does the word “temper” most nearly mean?

A Harden

B Moderate ✓ correct answer

C Abandon

D Anticipate

Why this answer. The newer approaches soften the older absolute view rather than discarding it — they still conserve, but accept some aging. 'Abandon' overstates it and 'harden' reverses it.

4

CAS / CAS-WV

HARD

Although the first printed maps of the region were widely copied, their coastlines were largely _____, drawn from sailors' recollections rather than from any survey.

Which choice completes the text with the most logical and precise word or phrase?

A conjectural ✓ correct answer

B official

C decorative

D restricted

Why this answer. Coastlines drawn from recollection rather than survey are guessed at. The 'although' concedes the maps' popularity before this limitation, and none of the other choices describes how a line was arrived at.

5

CAS / CAS-WV

HARD

Early mechanical clocks were notoriously ____: two clocks in the same town could disagree by a quarter of an hour by the end of a single day.

Which choice completes the text with the most logical and precise word or phrase?

A ornate

B portable

C expensive

D imprecise ✓ correct answer

Why this answer. The colon spells out the blank: clocks that disagree by a quarter of an hour keep inaccurate time. Ornateness, cost and portability are never at issue.

6

CAS / CAS-TS

HARD

The Federalist Papers are a collection of 85 essays written by Alexander Hamilton, John Jay, and James Madison. They were published pseudonymously in the *Independent Journal* and other New York newspapers in 1787–88 and argue that New Yorkers should vote to ratify the proposed United States Constitution. Though the authorship of most of the individual essays is certain, that of a few is in question: for instance, while No. 15, “The Insufficiency of the Present Confederation to Preserve the Union,” was surely penned by Hamilton, No. 52, “The House of Representatives,” may have been written by either Hamilton or Madison.

Which choice best describes the overall structure of the text?

A The text mentions a collection of essays and then points out something about these essays that is not completely known. ✓ correct answer

B The text summarizes an argument made in a collection of essays and then suggests that the essays’ authors didn’t unanimously agree with the argument.

C The text lists the authors of a collection of essays and then notes that some of the essays were written by one person, while others were written by two people.

D The text describes why a collection of essays is notable and then details its publication history.

Why this answer. The text introduces the collection, then turns on ‘that of a few is in question’ — uncertain authorship. C misreads that uncertainty as co-authorship, and B invents disagreement among the authors.

Though John Crowley, author of *Engine Summer*, is perhaps not as well known as the most commercially successful American writers of the past fifty years, influential figures have championed his work, including the poet James Merrill and the literary critic Harold Bloom. In his afterword to Crowley's book *Little, Big*, Bloom praises the novel's adroit blend of what playwright Friedrich Schiller termed the naive and sentimental modes—while Schiller thought works could be classified as either naive (seeking to describe reality) or sentimental (seeking to develop ideas), *Little, Big* demonstrates that a work can be both.

Which choice best states the main purpose of the text?

- A To argue that all writing must be classified as belonging to one of two categories
- B To compare the work of a writer with the work of a poet who admired him
- C To explain what inspired an author to write a particular work
- D To present a reason why a literary critic is impressed by a certain novel ✓ correct answer

Why this answer. The text builds to why Bloom praises *Little, Big*: it blends two modes Schiller treated as exclusive. A inverts the point, B never compares Merrill's poetry with Crowley's novel, and nothing addresses Crowley's inspiration.

The following text is from Mick Herron's 2023 novel *The Secret Hours*. The narrator is describing members of a government committee.

Finally, and adding much-needed gravitas, was Sir Winston Day, whose features seemed moulded to adorn a bust, or possibly a stamp, and whose forehead was so evidently bulging with grey matter that it would have been impertinent to inquire too closely into the actual achievements his half century of public service had produced. His recently published memoirs possibly cast light on this enigma, but given that such details were not provided until after the thirty-page mark, they might as well have remained state secrets.

Based on the text, which choice best describes Sir Winston Day?

- A He looks like a person worthy of respect, but his memoirs reveal that some of his actions were dishonorable.
- B He has maintained a modest profile even though he has served the public capably for many years.
- C He has the appearance of a distinguished figure, but it is uncertain whether he has accomplished anything to earn distinction. ✓ correct answer
- D He would be a celebrated public figure if his achievements did not have to be kept secret.

Why this answer. The look is settled — features fit for a bust, a forehead bulging with grey matter — while the achievements are pointedly unexamined ('impertinent to inquire too closely') and the memoirs bury them past page thirty. The narrator never says his actions were dishonorable (A) or secret (D).

9

INI / INI-CI

HARD

The following text is adapted from Jerome K. Jerome's 1889 novel *Three Men in a Boat (To Say Nothing of the Dog)*. The narrator is traveling by boat with Harris and another friend.

[Harris] told us anecdotes of how he had gone across the [English] Channel when it was so rough that the passengers had to be tied into their [beds], and he and the captain were the only two living souls on board who were not ill. Sometimes it was he and the second mate who were not ill; but it was generally he and one other man. If not he and another man, then it was he by himself.

Which choice best states the main idea of the text?

A During a previous boat trip, Harris spent more time with the other passengers than with the captain.

B Harris has a hard time remembering his first trip across the English Channel when his friends ask about it.

C When Harris speaks of an earlier trip, he often changes the details but always brags about his own wellness. ✓
correct answer

D Harris is worried that he and his friends will encounter rough waters during their boat trip.

Why this answer. The companion shifts each retelling — the captain, then the second mate, then 'one other man', then nobody — while the one fixed element is that Harris was never ill. That is a changing story with a constant boast, not forgetfulness (B).

10

INI / INI-CE

HARD

Male orangutans typically disperse from the territory in which they were born when they reach maturity, whereas females typically do not. Beatrice Ehmann and her colleagues showed that this difference in life trajectory is reflected in the diets of immature orangutans, with females' diets overlapping more with their mothers' diets than males' diets do. For example, on average, individual 9's mother shared approximately _____

Which choice most effectively uses data from the graph to complete the example?

A 16% of its diet with individual 9's diet, whereas individual 12's mother shared approximately 72% of its diet with individual 12's diet.

B 81% of its diet with individual 9's diet, whereas individual 12's mother shared approximately 72% of its diet with individual 12's diet. ✓ correct answer

C 81% of its diet with individual 9's diet, whereas individual 12's mother shared approximately 20% of its diet with individual 12's diet.

D 16% of its diet with individual 9's diet, whereas individual 12's mother shared approximately 20% of its diet with individual 12's diet.

Why this answer. Individual 9 is the female and 12 the male, and the claim is that females overlap more. The 'MD shared with ID' bars read about 81% for individual 9 and about 72% for individual 12, which is B. The 16% and 20% figures come from the other pair of bars, 'ID not shared with MD'.

11

INI / INI-CE

HARD

Sea otters eat large numbers of sea urchins, and sea urchins graze on kelp. Along stretches of coast where otters have returned after a long absence, kelp forests have expanded within a few years. A marine ecologist concluded that the otters' return is what allowed the kelp to recover, rather than any change in water temperature over the same period.

Which finding, if true, would most directly support the ecologist's conclusion?

A Kelp grows more slowly in warmer water than in colder water.

B Along stretches of coast where water temperature changed by the same amount but otters did not return, kelp forests did not expand. ✓ correct answer

C Sea urchin populations along the coast have declined over the past decade.

D Sea otters were absent from the region for several decades before their recent return.

Why this answer. The conclusion singles out the otters over temperature, so the support must hold temperature constant and vary the otters — which is what B does. C omits the cause of the urchin decline, and A and D restate background the passage already gives.

12

INI / INI-CE

HARD

Ecologist Mireia Solà has proposed that the reintroduction of beavers to a river catchment reduces the severity of downstream flooding, because the dams the animals build hold back water during heavy rain and release it slowly. Critics of the proposal note that beaver dams are small and frequently breached, and argue that any effect on a large catchment would be negligible. Solà is preparing a study to test her hypothesis.

Which finding from Solà's study, if true, would most directly support her hypothesis?

A Beaver dams in the catchment were rebuilt within days of being breached during storms.

B The number of beavers in the catchment increased steadily over the study period.

C Peak river levels downstream of the dammed reaches were substantially lower after reintroduction than before it, for storms of comparable rainfall. ✓ correct answer

D Water held behind the dams was clearer than water in undammed stretches of the river.

Why this answer. The hypothesis is specifically about downstream flood severity, so the supporting finding has to measure that, and has to hold rainfall comparable so the comparison means something — which is exactly C. A answers the critics' point about breaching but does not itself show any flood effect. B measures the population, not the flooding. D is about water quality, a different outcome entirely.

13

INI / INI-IE

HARD

Zooarchaeologist Ophélie Lebrasseur and her team examined a fox skeleton discovered in 1991 at an archaeological site alongside artifacts of human habitation (like spear points) in central Argentina. Lebrasseur et al. determined that the fox was *Dusicyon avus*, an extinct species resembling a jackal, and radiocarbon dating placed the fox at the site at the same time as human inhabitants. (Indeed, the inhabitants may have deliberately buried the fox.) In addition, while wild foxes have a diet entirely made of meat, isotopic signatures of the skeleton's teeth indicated that the fox's diet, like that of the humans, was partly composed of plant material. Lebrasseur et al. therefore concluded that _____

Which choice most logically completes the text?

- A the humans who were alive at the same time as the fox most likely ate more meat than the fox did.
- B the fox had a diet more similar to that of jackals than to that of wild foxes.
- C the humans who were alive at the same time as the fox hunted using the spears whose points were also found at the site.
- D the fox may have been a companion animal of the humans who inhabited the site at the same time. ✓ correct answer

Why this answer. Three strands converge on it: the fox lies among human artifacts, may have been deliberately buried, and ate plant material as the humans did rather than the all-meat diet of wild foxes. B invokes jackals, which the text mentions only as a resemblance in appearance.

14

INI / INI-IE

HARD

Biologists Rebecca M. Calisi-Rodriguez and George E. Bentley examined research on species such as dark-eyed juncos and tucos, which have both been studied under laboratory conditions as well as in the wild, to see whether there were significant differences between findings in the wild and in the lab. And, for example, they found for tucos that daytime is the most active period for wild individuals but not for captive individuals. Calisi-Rodriguez and Bentley therefore concluded that the laboratory setting was likely _____

Which choice most logically completes the text?

- A more suitable than wild settings for studying tucos' patterns of rest and activity.
- B affecting the results for both tucos and dark-eyed juncos.
- C more suitable for studying dark-eyed juncos than for studying tucos.
- D interfering with tucos' patterns of rest and activity. ✓ correct answer

Why this answer. Captive tucos lose the daytime activity peak that wild ones have, so captivity is changing the behaviour being measured. The text gives no junco result (B) and never compares suitability across species (C) or endorses the lab (A).

15

SEC / SEC-BS

HARD

Included in *Black Refractions: Highlights from The Studio Museum in Harlem*, a 2019 group exhibition at the Museum of the African Diaspora in San Francisco, California, was the work of artist Lorna Simpson, who is best known for her multimedia artworks that juxtapose images of African American women with text fragments. Her work challenges conventional notions of race, gender, history, and _____ she is credited with expanding the horizons of conceptual photographic art.

Which choice completes the text so that it conforms to the conventions of Standard English?

A memory, and ✓ correct answer

B memory and

C memory,

D memory

Why this answer. 'Memory' closes the list, and 'she is credited with...' is a second independent clause, so a comma plus 'and' is required. B omits the comma between independent clauses, C is a comma splice and D is a run-on.

16

SEC / SEC-BS

HARD

The practice of storing grain in raised stone granaries _____ several communities in the Atlas Mountains through centuries of unpredictable harvests.

Which choice completes the text so that it conforms to the conventions of Standard English?

A sustained ✓ correct answer

B sustained,

C ; sustained

D , sustained

Why this answer. 'The practice of storing grain in raised stone granaries' is the subject and 'sustained' its verb; nothing belongs between them.

17

SEC / SEC-BS

HARD

Sourdough bakers rely on a starter that is never entirely used _____ a portion is always held back to leaven the next loaf.

Which choice completes the text so that it conforms to the conventions of Standard English?

A up: ✓ correct answer

B up,

C up and

D up

Why this answer. The first clause is complete and the second explains what makes the starter never entirely used up, which is a colon's job. The comma alone splices two independent clauses, 'and' joins them without the comma it needs, and the unpunctuated version is a run-on.

18

SEC / SEC-BS

HARD

The Austronesian language family comprises some 1,200 languages—including the _____ Javanese and Bikol, which are spoken by 100 million and 4.6 million speakers, respectively—and accounts for one-fifth of the world's languages, making it of keen interest to linguists like Li Jen-kuei.

Which choice completes the text so that it conforms to the conventions of Standard English?

A languages ✓ correct answer

B languages:

C languages—

D languages,

Why this answer. A pair of dashes already sets off the whole 'including...' aside. Inside it, 'the languages Javanese and Bikol' is a noun plus its appositive names and takes no punctuation; a third dash or a colon would break the pair.

19

SEC / SEC-FS

HARD

The Akhundov National Library in Baku houses many historical newspapers, but it isn't necessary to travel to Baku to access them. Because these rare items are digitized, people around the world can access _____ regardless of where they live.

Which choice completes the text so that it conforms to the conventions of Standard English?

A it

B them ✓ correct answer

C this

D that

Why this answer. The antecedent is the plural 'these rare items' (the newspapers), so the plural pronoun 'them' is required.

20

SEC / SEC-FS

HARD

In 1783, the Laki fissure in Iceland began an eruption that lasted eight months. Ash carried on the wind _____ harvests as far away as central Europe that year.

Which choice completes the text so that it conforms to the conventions of Standard English?

A damages

B is damaging

C damaged ✓ correct answer

D will damage

Why this answer. 'In 1783' and 'that year' fix the event in the completed past, and the surrounding verbs ('began', 'lasted') are simple past.

21

SEC / SEC-FS

HARD

Because the glaciers of the Southern Alps advance and retreat on different schedules, surveyors measure _____ individually rather than treating the range as a single system.

Which choice completes the text so that it conforms to the conventions of Standard English?

A it

B that

C this

D them ✓ correct answer

Why this answer. The antecedent is the plural 'glaciers', so the plural object pronoun 'them' is required.

22

EOI / EOI-TR

HARD

Saffron must be harvested by hand, and each crocus flower yields only three usable stigmas. _____ a single kilogram of the spice represents the picking of roughly 150,000 flowers.

Which choice completes the text with the most logical transition?

A Nevertheless,

B In contrast,

C Similarly,

D Consequently, ✓ correct answer

Why this answer. The flower count follows from the hand harvest and the three-stigma yield, so the relation is causal rather than contrastive or parallel.

23

EOI / EOI-TR

HARD

Most bridges are designed so that the roadway carries the load down to the piers. _____ in a suspension bridge the roadway hangs from cables, and the load travels upward before it reaches the towers.

Which choice completes the text with the most logical transition?

A Likewise,

B For instance,

C By contrast, ✓ correct answer

D As a result,

Why this answer. An upward load path is the opposite of the downward one just described. 'Likewise' would require the two designs to agree, and a suspension bridge is not an instance of the general case.

24

EOI / EOI-TR

HARD

Green roofs reduce the volume of stormwater that reaches a city's drains, easing pressure on systems that were sized for a smaller population. _____ the soil layer insulates the building beneath it, lowering the energy needed to heat and cool the top floors.

Which choice completes the text with the most logical transition?

A In short,

B Instead,

C Moreover, ✓ correct answer

D Until then,

Why this answer. Insulation is a second, separate benefit added to the stormwater benefit. 'Instead' would replace the first claim and 'in short' would summarise it.

While researching a topic, a student has taken the following notes:

- The Clovis First hypothesis proposes that ancient humans known as Clovis were the first to populate North America.
- The Clovis arrived around 13,000 years ago.
- Stone tools found at the Saltville site in Virginia date to approximately 16,000 years ago.
- These items support the idea that humans were present in North America before the Clovis population arrived.
- More than a dozen North American sites containing pre-Clovis items have been identified.

The student wants to explain the implications of the discovery at the Saltville site. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A** It is the contention of the Clovis First hypothesis that the Clovis, an ancient human population, were the first to inhabit North America, arriving around 13,000 years ago.
- B** Located in Virginia, Saltville is one of more than a dozen North American sites where pre-Clovis items have been discovered.
- C** The approximately 16,000-year-old stone tools found at Saltville, a site in Virginia, should not be overlooked: they have important implications for the Clovis First hypothesis.
- D** Contrary to the Clovis First hypothesis, the approximately 16,000-year-old stone tools found at Saltville suggest that humans were in North America before the Clovis arrived. ✓ **correct answer**

Why this answer. Only D states what the discovery implies — humans predating the Clovis, against the hypothesis. C asserts that there are implications without naming them, A describes only the hypothesis, and B only locates the site.

While researching a topic, a student has taken the following notes:

- Cuneiform and hieroglyphic writing were both used in the ancient world.
- Cuneiform was written by pressing a reed stylus into wet clay.
- Hieroglyphs were most often carved into stone or painted onto plaster.
- Both systems were used for administrative records as well as religious texts.

The student wants to emphasize a difference between how the two scripts were produced. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A Cuneiform and hieroglyphic writing were both used for administrative records as well as religious texts.

B Cuneiform was pressed into wet clay with a reed stylus, whereas hieroglyphs were most often carved into stone or painted onto plaster. ✓ **correct answer**

C Cuneiform and hieroglyphic writing were two writing systems of the ancient world.

D Hieroglyphs served religious purposes in addition to administrative ones.

Why this answer. B names the production method for each and marks the contrast with 'whereas'. A and C state what the two share, and D describes only one script's uses.

While researching a topic, a student has taken the following notes:

- Minor planets are astronomical objects that orbit the Sun but are neither planets nor comets.
- Astronomer Jane Luu has discovered many minor planets in her career.
- Alongside colleagues David C. Jewitt and Chadwick Trujillo, she discovered the minor planet (19308) 1996 TO66 on October 12, 1996.
- Alongside colleagues David C. Jewitt, Chadwick Trujillo, and David J. Tholen, she discovered the minor planet (24978) 1998 HJ151 on April 28, 1998.

Which choice most effectively uses information from the given sentences to emphasize the discovery of (19308) 1996 TO66 to an audience already familiar with Jane Luu?

A First there was the discovery of (19308) 1996 TO66 on October 12, 1996, and then on April 28, 1998, Jane Luu discovered yet another minor planet with the help of her colleagues David C. Jewitt, Chadwick Trujillo, and David J. Tholen.

B Astronomer Jane Luu helped discover not only the minor planet (19308) 1996 TO66 on October 12, 1996, but also (24978) 1998 HJ151 on April 28, 1998.

C Jane Luu is an astronomer famous for her discovery of many minor planets in our solar system, including (19308) 1996 TO66.

D On October 12, 1996, Jane Luu and her colleagues David C. Jewitt and Chadwick Trujillo made the exciting discovery of the minor planet (19308) 1996 TO66. ✓ **correct answer**

Why this answer. The audience already knows who Luu is, so the sentence should stay on the one discovery. D does. C introduces her and treats TO66 as merely one of many, while A and B split attention across both discoveries.

Section 2, Module 1: Math

Standard · 22 questions · 35 minutes

1

ALG / ALG-LE

MEDIUM

A landscaping crew charges \$65 to visit a site plus \$28 for each hour worked. For one job the crew billed a total of \$289. For how many hours did the crew work?

A 4

B 6

C 8 ✓ correct answer

D 9

Derivation. $65 + 28h = 289$ gives $28h = 224$ and $h = 8$.

2

ALG / ALG-LE

MEDIUM

$$2x + 3y = 19$$

$$x - y = 2$$

For the given system of equations, what is the value of $x + y$?

A 5

B 6

C 8 ✓ correct answer

D 11

Derivation. $x = y + 2$ gives $5y + 4 = 19$, $y = 3$ and $x = 5$, so $x + y = 8$.

3

ALG / ALG-LF

MEDIUM

Line k passes through the points $(2, -1)$ and $(6, 7)$ in the xy -plane. What is the slope of line k ?

A $\frac{1}{2}$

B 2 ✓ correct answer

C 3

D 4

Derivation. $(7 - (-1)) / (6 - 2) = 8 / 4 = 2$.

4

ALG / ALG-LF

MEDIUM

A technician models the temperature of a cooling engine with $T(m) = 94 - 2.5m$, where T is the temperature in degrees Celsius m minutes after the engine is switched off. After how many minutes does the model predict a temperature of 34 degrees Celsius?

A 16

B 20

C 24 ✓ correct answer

D 38

Derivation. $94 - 2.5m = 34$ gives $2.5m = 60$ and $m = 24$.

5

ALG / ALG-LI

MEDIUM

Which of the following describes all values of x that satisfy $4 - 2x > 10$?

A $x > 3$ B $x < 3$ C $x > -3$ D $x < -3$ ✓ correct answer

Derivation. $-2x > 6$, and dividing by -2 reverses the sign: $x < -3$.

6

ALG / ALG-LE

MEDIUM

A caterer charges a fixed booking fee plus a fixed amount per guest. The total charge for 20 guests is \$930, and the total charge for 35 guests is \$1,470. What is the caterer's booking fee, in dollars?

A \$140

B \$170

C \$180

D \$210 ✓ correct answer

Derivation. Per guest = $(1470 - 930)/15 = 36$; fee = $930 - 20(36) = 210$.

7

ALG / ALG-LF

MEDIUM

A tank contains 240 litres of water and is drained at a constant rate of 8 litres per minute. Which equation gives the volume V , in litres, of water remaining in the tank t minutes after draining begins, for $0 \leq t \leq 30$?

A $V = 240 + 8t$

B $V = 8t - 240$

C $V = 240 - \frac{t}{8}$

D $V = 240 - 8t$ ✓ correct answer

Derivation. Starts at 240 and falls 8 per minute.

8

ADV / ADV-EQ

MEDIUM

A rectangular garden bed has a length of $2x - 5$ metres and a width of $x + 4$ metres. Which expression represents the area of the bed, in square metres?

A $3x - 1$

B $2x^2 + 3x - 20$ ✓ correct answer

C $2x^2 - 3x - 20$

D $2x^2 - 20$

Derivation. $(2x-5)(x+4) = 2x^2 + 3x - 20$.

9

ADV / ADV-NF

MEDIUM

A cinema finds that its daily profit, in dollars, is modelled by $P(d) = d^2 - 8d + 13$, where d is the ticket discount in dollars. What is the least daily profit predicted by this model?

A -3 ✓ correct answer

B -1

C 4

D 13

Derivation. Vertex at $d = 4$; $P(4) = 16 - 32 + 13 = -3$.

10

ADV / ADV-NE

MEDIUM

A colony of algae covers an area that triples each week. The colony covered 3 square metres at the start of the study and covers 243 square metres now. How many weeks have passed since the start of the study?

A 2

B 3

C 4 ✓ correct answer

D 27

Derivation. $3 \cdot 3^w = 243$ gives $3^w = 81$ and $w = 4$.

11

ADV / ADV-NE

MEDIUM

The two solutions to the equation $x^2 - 5x - 24 = 0$ are the lengths, in centimetres, of two segments in a construction diagram, one of which is discarded for being negative. What is the length of the segment that is kept?

A 3

B 5

C 8 ✓ correct answer

D 24

Derivation. Roots are 8 and -3; the positive one is 8.

12

ADV / ADV-EQ

MEDIUM

Which expression is equivalent to $\frac{12a^5b^3}{3a^2b^7}$ for $a > 0$ and $b > 0$?

A $\frac{4a^3}{b^4}$ ✓ correct answerB $4a^3b^4$ C $\frac{4a^7}{b^{10}}$ D $\frac{9a^3}{b^4}$

Derivation. $12/3 = 4$, $a^{(5-2)} = a^3$, $b^{(3-7)} = b^{-4}$.

13

ADV / ADV-NF

MEDIUM

A ball is thrown upward from a platform. Its height h , in metres, t seconds after release is given by $h(t) = -5t^2 + 20t + 12$. How many seconds after release does the ball reach its greatest height?

A 2 ✓ correct answer

B 4

C 12

D 32

Derivation. Vertex at $t = -20/(2(-5)) = 2$.

14

ADV / ADV-NF

MEDIUM

The population of a colony of bacteria triples every 6 hours. If the colony starts with 400 bacteria, which function gives the number of bacteria, $P(t)$, after t hours?

A $P(t) = 400(3)^{6t}$

B $P(t) = 400(3)^{\frac{t}{6}}$ ✓ correct answer

C $P(t) = 400(6)^{\frac{t}{3}}$

D $P(t) = 400 + 3t$

Derivation. Tripling every 6 hours means exponent $t/6$.

15

PSDA / PSDA-RP

MEDIUM

A recipe uses 3 cups of flour for every 4 cups of milk. If a baker uses 18 cups of flour, how many cups of milk are needed?

A 13.5

B 19

C 24 ✓ correct answer

D 27

Derivation. $18 \cdot \frac{4}{3} = 24$.

16

PSDA / PSDA-RP

MEDIUM

The price of a jacket was reduced by 20%, and the reduced price was then reduced by a further 15%. The final price is what percent of the original price?

A 65%

B 68% ✓ correct answer

C 70%

D 72%

Derivation. $0.80 \times 0.85 = 0.68$.

17

PSDA / PSDA-DI

MEDIUM

The table shows the number of books borrowed from a library on each of five days.

Day	Books borrowed
Monday	42
Tuesday	38
Wednesday	51
Thursday	47
Friday	62

What is the median number of books borrowed per day?

A 42

B 47 ✓ correct answer

C 48

D 51

Derivation. Sorted: 38, 42, 47, 51, 62; the middle value is 47.

18

PSDA / PSDA-ST

MEDIUM

A quality inspector examines a batch of 400 components and finds that 26 of them are defective. Based on this rate, how many defective components would be expected in a shipment of 5,000 components?

A 260

B 325 ✓ correct answer

C 400

D 520

Derivation. $26/400 * 5000 = 325$.

19

GT / GT-LA

MEDIUM

In a triangle, an exterior angle at one vertex measures 118° . One of the two non-adjacent interior angles measures 47° . What is the measure, in degrees, of the other non-adjacent interior angle?

A 47

B 62

C 71 ✓ correct answer

D 133

Derivation. An exterior angle equals the sum of the two remote interior angles: $118 - 47 = 71$.

20

GT / GT-AV

MEDIUM

student-produced response

A right circular cylinder has a radius of 5 centimetres and a height of 12 centimetres. What is the volume, in cubic centimetres, of the cylinder? (Round to the nearest whole number.)

Accepted answer(s): **942**

Derivation. $\pi * 25 * 12 = 300\pi = 942.48\dots$, which rounds to 942.

21

ALG / ALG-LE

MEDIUM

student-produced response

The sum of three consecutive even integers is 138. What is the greatest of the three integers?

Accepted answer(s): **48**

Derivation. $n + (n+2) + (n+4) = 138$ gives $n = 44$; the greatest is 48.

22

GT / GT-TR

MEDIUM

student-produced response

In right triangle PQR, angle Q is a right angle, $PQ = 9$ and $QR = 12$. What is the length of PR?

Accepted answer(s): **15**

Derivation. $\sqrt{81+144} = \sqrt{225} = 15$.

Section 2, Module 2: Math

Easy · 22 questions · 35 minutes

1

ALG / ALG-LE

EASY

If $\frac{n}{4} = 12$, what is the value of n ?

A 3

B 16

C 36

D 48 ✓ correct answer

Derivation. $n = 48$

2

ALG / ALG-LF

EASY

A pool contains 900 liters of water and is being drained at a constant rate of 30 liters per minute. Which equation gives the number of liters L remaining in the pool after m minutes of draining?

A $L=900+30m$

B $L=900-30m$ ✓ correct answer

C $L=30m-900$

D $L=900m-30$

Derivation. start 900, subtract 30 per minute

3

ALG / ALG-LF

EASY

What is the slope of the line that passes through the points (2,1) and (6,13) in the xy -plane?

A $\frac{1}{3}$

B 2

C 3 ✓ correct answer

D 7

Derivation. $(13-1)/(6-2) = 3$

4

ALG / ALG-LE

EASY

If $5(t+3)=2(t+3)+18$, what is the value of t ?

A 3 ✓ correct answer

B 5

C 6

D 9

Derivation. $3(t+3)=18 \rightarrow t+3=6 \rightarrow t=3$

5

ALG / ALG-LI

EASY

A student needs at least 240 points to pass a course and already has 165 points. Which inequality gives the possible numbers of additional points p the student needs?

A $p \leq 75$

B $p \geq 75$ ✓ correct answer

C $p \geq 405$

D $p \leq 240$

Derivation. $165 + p \geq 240 \rightarrow p \geq 75$

6

ALG / ALG-LE

EASY

$$5x+2y=31$$

$$5x-2y=9$$

What is the value of x in the solution to the given system of equations?

A 4 ✓ correct answer

B 5.5

C 8

D 11

Derivation. adding the equations gives $10x = 40 \rightarrow x = 4$

7

ALG / ALG-LF

EASY

student-produced response

A taxi ride costs \$3.50 plus \$2.00 for each mile traveled. How many miles were traveled on a ride that cost \$19.50?

Accepted answer(s): 8

Derivation. $(19.50 - 3.50) / 2 = 8$

8

ALG / ALG-LE

EASY

A number decreased by 14 is equal to 3 times the number. What is the number?

A -7 ✓ correct answer

B -3.5

C 3.5

D 7

Derivation. $x - 14 = 3x \rightarrow -14 = 2x \rightarrow x = -7$

9

ADV / ADV-EQ

EASY

Which expression is equivalent to $(2c^3)(7c^4)$?

A $9c^7$

B $14c^7$ ✓ correct answer

C $14c^{12}$

D $9c^{12}$

Derivation. $2 \cdot 7 = 14$, $3+4 = 7$

10

ADV / ADV-EQ

EASY

Which expression is equivalent to $\frac{9x+18}{3}$?

A $3x+6$ ✓ correct answer

B $3x+18$

C $9x+6$

D $3x + \frac{18}{3}x$

Derivation. dividing each term by 3 gives $3x + 6$

11

ADV / ADV-NF

EASY

The function f is defined by $f(x) = \frac{24}{x}$. What is the value of $f(6)$?

A 4 ✓ correct answer

B 18

C 30

D 144

Derivation. $24 / 6 = 4$

12

ADV / ADV-NE

EASY

What is the smaller of the two solutions to the equation $(x-8)(x+2)=0$?

A -8

B -2 ✓ correct answer

C 2

D 8

Derivation. solutions are 8 and -2; the smaller is -2

13

ADV / ADV-NE

EASY

If $\sqrt{2y} = 8$, what is the value of y ?

A 4

B 16

C 32 ✓ correct answer

D 64

Derivation. $2y = 64 \rightarrow y = 32$

14

ADV / ADV-NF

EASY

student-produced response

What is the value of $\sqrt{169} + \sqrt{36}$?

Accepted answer(s): **19**

Derivation. $13 + 6 = 19$

15

PSDA / PSDA-RP

EASY

In a survey, 18 of the 300 people surveyed said they had never used a public library. What percentage of the people surveyed said they had never used a public library?

A 6% ✓ correct answer**B** 12%**C** 18%**D** 60%**Derivation.** $18/300 = 0.06$

16

PSDA / PSDA-RP

EASY

A map uses a scale in which 1 centimeter represents 25 kilometers. Two cities are 7 centimeters apart on the map. What is the actual distance, in kilometers, between the two cities?

A 32**B** 125**C** 175 ✓ correct answer**D** 250**Derivation.** $7 \times 25 = 175$

17

PSDA / PSDA-DI

EASY

TABLE_B The table shows the number of tickets sold for each of four film screenings. How many more tickets were sold for the screening with the greatest number of tickets sold than for the screening with the least?

A 43**B** 79**C** 92 ✓ correct answer**D** 137**Derivation.** $137 - 45 = 92$

18

PSDA / PSDA-ST

EASY

The mean of four numbers is 15. Three of the numbers are 9, 14, and 18. What is the fourth number?

A 15

B 19 ✓ correct answer

C 22

D 60

Derivation. total 60; $60 - (9+14+18) = 19$

19

GT / GT-LA

EASY

In a triangle, the measures of two angles are 38° and 61° . What is the measure, in degrees, of the third angle?

A 71

B 81 ✓ correct answer

C 99

D 109

Derivation. $180 - 38 - 61 = 81$

20

GT / GT-AV

EASY

A rectangular prism has a length of 5 meters, a width of 4 meters, and a height of 3 meters. What is the volume, in cubic meters, of the prism?

A 12

B 24

C 60 ✓ correct answer

D 94

Derivation. $5 \times 4 \times 3 = 60$

21

GT / GT-AV

EASY

A parallelogram has a base of 12 inches and a height of 9 inches. What is the area, in square inches, of the parallelogram?

A 21

B 42

C 54

D 108 ✓ correct answer

Derivation. base * height = $12 \cdot 9 = 108$

22

GT / GT-TR

EASY

student-produced response

In right triangle DEF, angle F is a right angle, $DE=17$, and $EF=8$. What is the length of DF?

Accepted answer(s): **15**

Derivation. $\sqrt{289-64} = 15$

Section 2, Module 2: Math

Hard · 22 questions · 35 minutes

1

ALG / ALG-LE

HARD

A theatre sold 340 tickets for a performance. Adult tickets cost \$18 and student tickets cost \$11, and the total revenue was \$4,790. How many student tickets were sold?

A 130

B 180

C 190 ✓ correct answer

D 210

Derivation. $18a + 11s = 4790$ with $a + s = 340$ gives $7a = 4790 - 3740 = 1050$, $a = 150$, $s = 190$.

2

ALG / ALG-LE

HARD

$$9x - 3y = 15$$

$$cx - y = 8$$

In the given system of equations, c is a constant. For which value of c does the system have no solution?

A -3

B 3 ✓ correct answer

C 8

D 9

Derivation. Parallel needs $c/9 = 1/3$, so $c = 3$, and the constants then disagree.

3

ALG / ALG-LF

HARD

A line in the xy -plane passes through $(-4, 9)$ and is perpendicular to the line $y = \frac{2}{3}x + 5$. What is the y -coordinate of the point where the line crosses the y -axis?

A -3

B 1

C 3 ✓ correct answer

D 15

Derivation. Perpendicular slope is $-3/2$; $9 = -3/2(-4) + b$ gives $b = 3$.

4

ALG / ALG-LI

HARD

A shop sells notebooks for \$4 each and folders for \$3 each. A customer buys a total of 20 items and spends no more than \$68. What is the greatest number of notebooks the customer could have bought?

A 6

B 7

C 8 ✓ correct answer

D 12

Derivation. $4n + 3(20-n) \leq 68$ gives $n \leq 8$.

5

ADV / ADV-NE

HARD

$$x^2 + bx + 45 = 0$$

In the given equation, b is a constant. If the equation has exactly one real solution, and b is positive, what is the value of b ?

A $3\sqrt{5}$

B 9

C $6\sqrt{5}$ ✓ correct answer

D 45

Derivation. Discriminant $b^2 - 180 = 0$ gives $b = 6\sqrt{5}$ for the positive root.

6

ADV / ADV-NF

HARD

The function g is defined by $g(x) = 2x^2 - 12x + 7$. The graph of $y = g(x)$ in the xy -plane has its vertex at the point (h, k) . What is the value of $h + k$?

A -11

B -8 ✓ correct answer

C -5

D 3

Derivation. $h = 3$, $k = g(3) = 18 - 36 + 7 = -11$, so $h + k = -8$.

7

ADV / ADV-EQ

HARD

The functions f and g are defined by $f(x) = x^2 - 1$ and $g(x) = 2x + 3$. What is the value of $f(g(2))$?

A 15

B 24

C 48 ✓ correct answer

D 49

Derivation. $g(2) = 7$ and $f(7) = 49 - 1 = 48$.

8

ADV / ADV-NE

HARD

A cable of length $x + 4$ metres is stretched so that its length equals $\sqrt{2x + 11}$ metres. Given that the length is positive, what is the value of x ?

A -5

B -1 ✓ correct answer

C 1

D 5

Derivation. $2x+11 = (x+4)^2$ gives $x^2+6x+5 = 0$; of the roots -1 and -5 only -1 keeps the length positive.

9

ADV / ADV-NF

HARD

The function f is defined by $f(x) = k(2)^x$, where k is a positive constant. If $f(5) - f(3) = 96$, what is the value of k ?

A 4 ✓ correct answer

B 6

C 8

D 12

Derivation. $k(32-8) = 24k = 96$, so $k = 4$.

10

ADV / ADV-EQ

HARD

The side length of a cube, in centimetres, is $\sqrt[3]{V}$, where V is its volume in cubic centimetres. If a cube has volume $64y^{12}$ cubic centimetres, where $y > 0$, what is its side length, in centimetres?

A $4y^4$ ✓ correct answer

B $8y^4$

C $4y^9$

D $16y^4$

Derivation. $64^{(1/3)} = 4$ and $(y^{12})^{(1/3)} = y^4$.

11

ADV / ADV-NF

HARD

The function h is defined by $h(x) = 3x^2 + 6x - 2$. For what value of x does $h(x)$ attain its minimum, and what is that minimum value?

A $x = -1$, minimum -5 ✓ correct answer

B $x = -1$, minimum 1

C $x = 1$, minimum 7

D $x = -2$, minimum -2

Derivation. Vertex $x = -1$; $h(-1) = 3 - 6 - 2 = -5$.

12

PSDA / PSDA-ST

HARD

A data set of 8 values has a mean of 24. When a ninth value is added, the mean of the 9 values is 26. What is the ninth value?

A 26

B 34

C 42 ✓ correct answer

D 50

Derivation. $9(26) - 8(24) = 234 - 192 = 42$.

13

PSDA / PSDA-ST

HARD

The table summarises the results of a survey of 200 students.

	Prefers cycling	Prefers walking	Total
Year 10	48	52	100
Year 11	62	38	100
Total	110	90	200

One of the students who prefers cycling is selected at random. What is the probability that the student is in Year 11?

A $\frac{31}{100}$

B $\frac{31}{55}$ ✓ correct answer

C $\frac{11}{20}$

D $\frac{62}{90}$

Derivation. $62/110 = 31/55$.

14

PSDA / PSDA-RP

HARD

A car travels the first 60 kilometres of a journey at an average speed of 40 kilometres per hour and the remaining 60 kilometres at an average speed of 60 kilometres per hour. What is the car's average speed, in kilometres per hour, for the whole journey?

A 44

B 48 ✓ correct answer

C 50

D 52

Derivation. Total time $1.5+1 = 2.5$ h for 120 km, so 48 km/h.

15

PSDA / PSDA-DI

HARD

A biologist measured the mass, in grams, of each of 40 seedlings. The mean mass was reported as 6.4 grams. The biologist later found that one measurement had been recorded as 9.2 grams when the true value was 2.2 grams. What is the correct mean mass, in grams?

A 6.05

B 6.225 ✓ correct answer

C 6.4

D 6.575

Derivation. Total falls by 7, so the mean falls by $7/40 = 0.175$ to 6.225.

16

GT / GT-LA

HARD

In triangle DEF, side DE has length 12 and side DF has length 18. Point G lies on DE and point H lies on DF so that GH is parallel to EF. If $DG = 8$, what is the length of DH?

A 10

B 12 ✓ correct answer

C 13.5

D 16

Derivation. $DG/DE = DH/DF$ gives $8/12 = DH/18$, so $DH = 12$.

17

GT / GT-AV

HARD

A sphere and a right circular cone have equal volumes. The sphere has radius 3 and the cone has radius 3. What is the height of the cone?

A 4

B 9

C 12 ✓ correct answer

D 36

Derivation. $(4/3)\pi(27) = (1/3)\pi(9)h$ gives $36\pi = 3\pi h$, $h = 12$.

18

GT / GT-TR

HARD

A ramp rises 5 metres over a slope length of 13 metres. What is the horizontal distance, in metres, covered by the ramp?

A 8

B 12 ✓ correct answer

C $\sqrt{194}$

D 18

Derivation. $\sqrt{13^2 - 5^2} = \sqrt{144} = 12$.

19

ALG / ALG-LF

HARD

A subscription service charges the same amount each month. A customer who has paid for 2 months has been billed \$47 in total, including a one-time joining fee, and a customer who has paid for 6 months has been billed \$131 in total. What is the one-time joining fee, in dollars?

A \$5 ✓ correct answer

B \$11

C \$21

D \$26

Derivation. Monthly = $(131-47)/4 = 21$; fee = $47 - 2(21) = 5$.

20

ADV / ADV-NE

HARD

student-produced response

$$2x^2 - 11x + 12 = 0$$

What is the sum of the solutions to the given equation? (Express your answer as a fraction or a decimal.)

Accepted answer(s): **11/2 or 5.5**

Derivation. Sum of roots = $-b/a = 11/2$.

21

ALG / ALG-LE

HARD

student-produced response

A chemist has one solution that is 20% acid and another that is 50% acid. How many litres of the 50% solution must be added to 12 litres of the 20% solution to produce a mixture that is 30% acid?

Accepted answer(s): **6**

Derivation. $0.2(12) + 0.5v = 0.3(12+v)$ gives $2.4 + 0.5v = 3.6 + 0.3v$, $v = 6$.

22

GT / GT-AV

HARD

student-produced response

A rectangular prism has a square base. Its height is 3 times the side length of the base, and its volume is 375 cubic centimetres. What is the side length, in centimetres, of the base?

Accepted answer(s): 5

Derivation. $s^2 \cdot 3s = 3s^3 = 375$ gives $s^3 = 125$, $s = 5$.